



## Asset Performance: Ene 07 Internal lighting



Contributes to 40 credits  
available for asset energy  
rating

No Minimum Standard

### Aim

To minimise operational energy consumption and the associated carbon emissions by increasing the intrinsic energy efficiency of the building fabric and the installed buildings services.

### Question

1. What percentage of lit asset floor area is fitted with LED lamps?
  - a. What percentage of LED lighting is fitted with constant illuminance controls?
2. What percentage of lit asset floor area is fitted with Compact Fluorescent lamps?
  - a. What percentage of Compact Fluorescent lamps have diffusers or shades fitted?
3. What percentage of lit asset floor area is fitted with Tungsten Halogen lamps?
  - a. What percentage of Tungsten Halogen lamps have diffusers or shades fitted?
4. What percentage of lit asset floor area is fitted with Incandescent lamps?
  - a. What percentage of Incandescent lamps have diffusers or shades fitted?
5. What percentage of lit asset floor area is fitted with T12 fluorescent tubes?
  - a. What percentage of T12 fluorescent tubes have diffusers or shades fitted?
6. What percentage of lit asset floor area is fitted with T8 fluorescent tubes?
  - a. What percentage of T8 fluorescent tubes have diffusers or shades fitted?
7. What percentage of lit asset floor area is fitted with T5 fluorescent tubes?
  - a. What percentage of T5 fluorescent tubes have diffusers or shades fitted?
8. What percentage fluorescent tubes are fitted with constant illuminance controls?
9. What percentage of fluorescent tubes are fitted with high frequency ballasts?
10. What percentage of lit asset floor area is fitted with metal halide lamps?
  - a. What percentage of metal halide lamps have diffusers or shades fitted?
11. What percentage of lit asset floor area has access to daylight?
  - a. What percentage of day lit space is fitted with dimmable photoelectric control?
  - b. What percentage of day lit space is fitted with photoelectric switching controls?
12. What percentage of the lit asset floor area is occasionally occupied?

- a. What percentage of occasionally occupied space is fitted with auto off (manual on) presence detection?
- b. What percentage of occasionally occupied space is fitted with auto on and off presence detection?

| Credits | Question No. | Question                                                                    | Answer | a.     | b.     |
|---------|--------------|-----------------------------------------------------------------------------|--------|--------|--------|
|         |              |                                                                             |        | Answer | Answer |
| -       | 1.           | LED lighting                                                                | %      | %      | n/a    |
| -       | 2.           | Compact Fluorescent lamps                                                   | %      | %      | n/a    |
| -       | 3.           | Tungsten Halogen lamps                                                      | %      | %      | n/a    |
| -       | 4.           | Incandescent lamps                                                          | %      | %      | n/a    |
| -       | 5.           | T12 fluorescent tubes                                                       | %      | %      | n/a    |
| -       | 6.           | T8 fluorescent tubes                                                        | %      | %      | n/a    |
| -       | 7.           | T5 fluorescent tubes                                                        | %      | %      | n/a    |
| -       | 8.           | Fluorescent tubes with constant illuminance controls                        | %      | n/a    | n/a    |
| -       | 9.           | Fluorescent tubes with high frequency ballasts                              | %      | n/a    | n/a    |
| -       | 10.          | Metal halide lamps                                                          | %      | %      | n/a    |
| -       | 11.          | What is the percentage of the lit asset floor area with access to daylight? | %      | %      | %      |
| -       | 12.          | What percentage of the lit asset floor area is occasionally occupied?       | %      | %      | %      |

## Methodology

### Percentage of asset floor area lit by lamp type

The sum of the percentage of lit asset floor area for all lamp types must equal 100%.

### Floor area calculations

The total floor area for all the fitted lamp types must equal 100%. Questions 1 - 7 and 10.

The total floor area of day lit space fitted controls must not exceed 100% of the daylit space. Questions 11a and 11b.

The total floor area of occasionally occupied space must not exceed 100% of occasionally occupied space. Questions 12a and 12b.

## Evidence

| Criteria | Evidence requirement                                                                                                                                                                                             |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -        | The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance. |
| All      | Photographic evidence of visual inspection of lighting types identified.                                                                                                                                         |
| All      | Copy of the building plans highlighting areas which use the mentioned lighting types, daylight and occupancy sensors for lighting.                                                                               |
| All      | Evidence to demonstrate how the percentage was calculated.                                                                                                                                                       |

## Definitions

### Diffuser or shade:

A diffuser or a shade is a transparent or translucent optical element that scatters in all directions the light emitted by the light source(s) contained within a luminaire.

### For dimmable lighting with constant illuminance control:

By implication constant illuminance control is intended to mean a lighting control scheme that provides constant illuminance on the task by dimming the light sources early in their life cycle when they have a higher output. The savings achievable through constant illuminance control are more significant for LED or fluorescent lighting compared to other lamp types because the percentage light output drops more significantly over the lifetime of the light source. Examples of constant illuminance controls include, but are not limited to, dimming ballasts or integral LED drivers used in conjunction with photodetectors to meet task illuminance requirements by controlling the light output of the lighting system.

### Occasionally occupied space:

Examples of occasionally occupied space includes toilets, kitchenette for an office, and storage areas. Occupied spaces can vary depending on the asset type.

## Additional information

### Asset energy calculator

In order to obtain the maximum possible number of credits for the asset energy performance as much information as possible should be entered. Where no value is entered, the asset energy rating calculator will assume a pessimistic default value.